Compression Profiles

A Signal-Class Architecture of
Differential Psychological Resolution

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Series: Structural Regulation Framework, Paper 4

Abstract

The Structural Regulation Framework (SRF; Halmetoja, 2026a) models how psychological regulation operates through interacting parameters: Internal Regulation Capacity, Structural Flexibility, External Regulation Dependence, Regulation Externalization Latency, and Externalization Vectors. However, SRF does not fully explain why the same individual may display radically different regulatory sophistication depending on the type of psychological signal being processed.

The present paper proposes that psychological regulation is not globally scalar. The mind does not process all psychologically meaningful signals at equal representational resolution. Instead, regulation operates in a signal-class-dependent manner, with different signal classes sustaining different effective resolution levels depending on activation conditions and structural organization.

The paper introduces four central constructs: (1) Signal Class — a psychologically meaningful category of experience requiring interpretation and regulation; (2) Representational Resolution — the degree to which a signal can be affectively, interpretively, and symbolically differentiated; (3) Compression — defensive reduction in representational resolution under activation; and (4) Compression Profile — the characteristic distribution of effective resolution across canonical signal classes.

A provisional signal taxonomy is proposed, organized into relational regulation (attachment, dependency, rejection, separateness), self-evaluative regulation (shame, status), and interpretive/control regulation (control/agency, intent attribution). The framework further proposes dynamic modulation: resolution varies between baseline and stress conditions, with selective degradation of vulnerable channels under activation.

This paper is conceptual and presents no original empirical data. Its contribution lies in proposing a formal structural model of differential representational resolution that generates testable predictions regarding within-subject variation, stress-selective degradation, characteristic clinical compression signatures, and channel-specific empathy failure.

Keywords: psychological resolution; signal processing; personality structure; defensive processes; emotion regulation; differential functioning; compression; mentalization

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1. Introduction

1.1 The Uniformity Problem

Many influential psychological models rely substantially on broad global constructs. A person is described as having high or low distress tolerance, strong or weak mentalization, secure or insecure attachment. While contemporary research increasingly acknowledges context-sensitivity, the dominant assessment and theoretical frameworks still operate primarily at the level of global parameters.

Clinical observation often challenges strong assumptions of regulatory uniformity.

The same individual may demonstrate extraordinary nuance in one domain while collapsing into minimally differentiated categorical processing in another. A person may sustain criticism in professional contexts while fragmenting under romantic uncertainty. A person may reason strategically about institutions while becoming categorically reactive to shame. A person may display sophisticated empathy in low-threat interactions while showing empathic failure under attachment activation.

These observations are not anomalies. They are pervasive. And they suggest that regulatory capacity is not globally scalar but structurally differentiated.

1.2 What SRF Does Not Explain

The Structural Regulation Framework (Halmetoja, 2026a) provides a mechanism-level account of regulation: how holding capacity (IRC), processing flexibility (SF), external dependence (ERD), externalization timing (REL), and vector direction (EV) interact to produce regulatory behavior. A companion paper on Resolution Resistance (Halmetoja, 2026b) further proposed that regulatory parameters may vary across domains, with defensive compression maintained preferentially in high-activation contexts.

However, neither paper fully formalizes the structural architecture of this variation. The domain-selective hypothesis (Halmetoja, 2026b, H4) identifies the phenomenon but does not provide a systematic framework for characterizing which signals compress, how compression distributes across signal types, or how compression profiles relate to personality organization.

The present paper addresses this gap.

1.3 The Central Question

The relevant question is not:

> "How psychologically flexible is this person?"

The relevant question is:

> "Which signal classes compress, and under what conditions?"

This reframing shifts personality assessment from global trait description toward structural topology — mapping the architecture of differential resolution across psychologically meaningful signal categories.

1.4 Scope

This paper proposes a conceptual architecture. It does not present empirical validation, validated instruments, or computational models. Its contribution lies in formalizing a structural framework that generates testable predictions and connects to established research traditions without claiming to replace them.

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2. Central Constructs

2.1 Signal Class

A signal class is defined as a psychologically meaningful category of experience requiring interpretation and regulation.

Signal classes are not contextual domains ("work," "relationship"). They are structural signal families — categories of internally relevant information that the regulatory system must process regardless of external context.

Candidate signal classes include:

- Rejection
- Shame
- Dependency
- Status threat
- Relational ambiguity
- Attachment availability
- Separateness
- Control/agency
- Intent attribution

Signal classes are derived provisionally from established research literatures: attachment theory (Bowlby, 1969; Mikulincer & Shaver, 2007), rejection sensitivity (Downey & Feldman, 1996), self-conscious emotions (Tangney & Dearing, 2002), social rank theory (Gilbert, 1997), agency and control research (Bandura, 1997), mentalization (Fonagy et al., 2002), and self–other differentiation (Bowen, 1978).

The taxonomy is intentionally bounded for explanatory tractability. It is not proposed as exhaustive ontology.

2.2 Representational Resolution

Representational resolution refers to the degree to which a signal can be:

- **Affectively differentiated**: experienced as nuanced emotional texture rather than undifferentiated arousal
- **Interpretively differentiated**: understood through multiple possible meanings rather than collapsed into a single categorical interpretation
- **Symbolically articulated**: represented in language, narrative, or reflective thought rather than remaining pre-symbolic

High resolution example: > "My partner is distant, but may simply be stressed. I notice my anxiety but can hold uncertainty."

Low resolution example: > "Distance means abandonment."

The term "representational resolution" is preferred over metaphorical alternatives (e.g., "bit depth") for formal use. The signal processing analogy remains heuristic — it illuminates structural similarities but does not claim literal isomorphism between psychological and digital systems.

Representational resolution is treated here as a higher-order composite construct rather than three independent dimensions. While affective differentiation, interpretive differentiation, and symbolic articulation may prove empirically separable, the present framework proposes them as facets of a single underlying capacity: the degree to which a given signal class can be processed with nuance rather than collapsed into categorical encoding. Whether this composite structure holds empirically remains an open measurement question.

2.3 Compression

Compression refers to defensive reduction in representational resolution under activation.

Three properties distinguish compression from other forms of cognitive simplification:

1. **Compression is not random.** It does not reduce resolution uniformly across all signals. It targets specific signal classes.
2. **Compression is structural.** It reflects stable architectural features of the regulatory system, not merely momentary cognitive load effects.
3. **Compression is activation-dependent.** It intensifies when full-resolution processing becomes destabilizing — when the regulatory cost of accurate representation exceeds available holding capacity for that signal class.

Compression occurs under conditions including:

- Shame overload
- Dependency exposure
- Control loss
- Rejection threat
- Attachment ambiguity under insufficient holding capacity

Compression is not inherently pathological. All psychological systems compress under sufficient activation. The clinically relevant question concerns which signals compress earliest, most severely, and most persistently.

2.4 Compression Profile

A compression profile is defined as the characteristic distribution of effective representational resolution across canonical signal classes.

Example (illustrative, not empirical):

Signal Class	Effective Resolution
Attachment	Low
Dependency	Very low
Rejection	Low
Shame	Very low
Status	High
Control	Moderate
Intent Attribution	Moderate-high
Separateness	Low

This profile describes a system that processes status and strategic information at high resolution while maintaining compressed processing of shame, dependency, rejection, and separateness signals.

The compression profile explains paradoxical functioning without requiring global dysfunction. A person may be sophisticated in one channel and minimally differentiated in another — not because of inconsistency, but because different signal classes invoke different effective resolution levels.

3. Canonical Signal Taxonomy

The following taxonomy is proposed as a provisional framework for organizing psychologically meaningful signal classes. It is intentionally bounded and empirically refinable.

3.1 Relational Regulation

Attachment

Organizing question: Are important others emotionally available?

This signal class concerns the perceived availability, responsiveness, and reliability of attachment figures. It draws on attachment theory (Bowlby, 1969; Ainsworth et al., 1978) and adult attachment research (Mikulincer & Shaver, 2007).

Dependency

Organizing question: Do I require external regulation to remain coherent?

This signal class concerns the system's recognition of its own structural reliance on external stabilization — specifically, the degree to which self-coherence depends on interpersonal regulatory input. It is distinct from attachment (which concerns perceived availability of others) and from helplessness (which concerns agency). Dependency in this framework refers narrowly to regulatory outsourcing: the felt necessity of external input for internal stability maintenance.

It draws on dependency research (Bornstein, 1992), self-regulation theory, and the ERD construct from SRF (Halmetoja, 2026a).

Rejection

Organizing question: Am I being excluded or abandoned?

This signal class concerns perceived social exclusion, abandonment, or relational termination. It draws on rejection sensitivity research (Downey & Feldman, 1996) and social pain literature (Eisenberger, 2012).

Separateness / Boundary

Organizing question: Where do I end and the other begin?

This signal class concerns self–other differentiation, individuation, and the capacity to tolerate the other's autonomy without experiencing it as threat. It draws on differentiation of self research (Bowen, 1978; Skowron & Friedlander, 1998).

3.2 Self-Evaluative Regulation

Shame

Organizing question: Am I diminished, defective, or exposed?

This signal class concerns self-evaluative affect that threatens the organization of the self rather than merely producing unpleasant experience. It draws on shame research (Lewis, 1971; Tangney & Dearing, 2002) and self-conscious emotion literature.

Status

Organizing question: Where do I stand relative to others?

This signal class concerns perceived social rank, comparative worth, and hierarchical positioning. It draws on social rank theory (Gilbert, 1997) and social comparison research.

3.3 Interpretive / Control Regulation

Control / Agency

Organizing question: Can I meaningfully influence outcomes?

This signal class concerns perceived capacity to affect the environment, predict consequences, and maintain effective agency. It draws on control theory (Carver & Scheier, 1998) and learned helplessness research.

Intent Attribution

Organizing question: What does the other person mean?

This signal class concerns the interpretation of others' motives and intentions specifically under conditions of ambiguity or potential threat. It is narrower than general theory of mind or mentalization — it refers specifically to the resolution at which ambiguous interpersonal behavior is interpreted when self-relevant stakes are present. A person may possess intact theory of mind in neutral contexts while compressing intent attribution to hostile/benign binaries under relational activation.

It draws on mentalization research (Fonagy et al., 2002), hostile attribution bias literature (Dodge, 1980), and epistemic trust research (Fonagy & Allison, 2014).

3.4 Taxonomic Constraints

This taxonomy is:

- **Provisional**: empirically refinable through factor-analytic investigation
- **Bounded**: intentionally limited to eight classes for explanatory tractability
- **Not exhaustive**: additional signal classes (loss/grief, moral evaluation, sexuality, disgust) may warrant future inclusion
- **Not mutually exclusive**: signal classes may co-activate and interact

Selection rationale: The eight proposed classes were selected because each (a) has an established independent research literature, (b) is clinically recognizable as a distinct source of regulatory activation, (c) produces characteristic defensive responses when compressed, and (d) can be differentially activated in experimental paradigms. Classes that met fewer of these criteria (e.g., grief, moral evaluation) were excluded from the initial taxonomy but may be incorporated in future revisions.

Orthogonality: The proposed signal classes are not assumed to be strictly orthogonal. Attachment, rejection, dependency, and separateness are conceptually adjacent and likely share variance. The taxonomy proposes them as partially overlapping regulatory dimensions rather than strictly independent constructs. Whether they prove empirically separable or collapse into fewer latent factors is an open question requiring factor-analytic investigation. The present taxonomy prioritizes clinical recognizability and differential activation potential over strict statistical independence.

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4. Dynamic Modulation

4.1 Static Profiles Are Insufficient

A compression profile is not a fixed trait. Resolution changes under activation. The same person may process attachment signals at moderate resolution under low-threat conditions while collapsing to minimal resolution under high-threat activation.

This dynamic property is essential to the framework's explanatory power.

4.2 Baseline Resolution Profile

The baseline resolution profile describes effective representational resolution under normal, low-threat functioning conditions.

Example (illustrative):

Signal Class	Baseline Resolution
Shame	Moderate
Attachment	Moderate-high
Status	High
Control	High

4.3 Stress Resolution Profile

The stress resolution profile describes effective representational resolution under activation — when the system is under regulatory load.

Example (same individual under activation):

Signal Class	Stress Resolution
Shame	Very low
Attachment	Low
Status	High
Control	Moderate-high

This explains state-dependent collapse: the same person who processes shame at moderate resolution under calm conditions may collapse to highly compressed categorical processing under shame activation.

4.4 Selective Degradation

The critical observation is that degradation under stress is not uniform. Some channels degrade severely while others remain relatively preserved. This selective degradation pattern is itself a structural feature of the system — it reflects which signal classes are most costly to process at full resolution.

4.5 Resolution Retention (Optional Construct)

Resolution retention refers to how much representational resolution survives under stress for a given signal class.

Example: - Baseline shame resolution: Moderate (5/8) - Stress shame resolution: Very low (1/8) - Resolution retention: 1/5 = 20%

This construct captures degradation magnitude. Signal classes with low resolution retention are the system's most vulnerable channels — the points where defensive compression activates earliest and most severely.

Whether resolution retention proves empirically useful as a distinct construct or is better captured by the baseline-stress difference alone remains an open question.

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5. Compression as Structure

5.1 Not Cognitive Load

Compression in the present framework is not equivalent to cognitive load effects. Cognitive load reduces processing capacity globally and temporarily. Compression is:

- **Signal-class-specific**: it targets particular experiential categories
- **Structurally stable**: it reflects architectural features that persist across situations
- **Defensively motivated**: it occurs because full-resolution processing is destabilizing, not merely because resources are depleted

A person under cognitive load may process all signals less effectively. A person with structural compression processes specific signals less effectively regardless of available cognitive resources.

5.2 Not Alexithymia

Alexithymia describes global difficulty identifying and describing emotions. Compression is signal-class-specific. A person may articulate emotions with extraordinary precision in some domains while remaining minimally differentiated in others.

5.3 Not Low Intelligence

Compression is proposed as conceptually distinct from general cognitive ability. Highly intelligent individuals may maintain severe compression in specific signal classes. Professional competence, strategic reasoning, and abstract thinking may coexist with minimally differentiated processing of shame, dependency, or attachment signals.

This distinction is clinically important because it explains why intelligence-based interventions (insight, psychoeducation, cognitive reframing) may fail to resolve structural compression. The limitation is not cognitive — it is regulatory.

5.4 Compression as Regulatory Architecture

Compression is proposed as a structural feature of regulatory organization — not a momentary state, not a cognitive limitation, and not a motivational deficit. It reflects the system's characteristic way of managing signals whose full-resolution processing would exceed available holding capacity.

This framing connects directly to the Resolution Resistance model (Halmetoja, 2026b): compression persists because it is structurally stabilizing. Decompression would require holding capacity that the system may not possess for that specific signal class.

6. Relationship to SRF

6.1 Extension, Not Replacement

The present framework does not replace SRF. It extends it by specifying where regulatory parameters operate differentially.

SRF explains why regulatory behavior occurs — through the interaction of IRC, SF, ERD, REL, and EV.

Signal-Class Architecture explains where regulatory collapse occurs — which signal classes lose resolution under which conditions.

6.2 Structural Mapping

At the most general level, the relationship can be understood as:

- **Compression Profile** = the observable regulatory interface (what the system does with different signals)
- **SRF** = the underlying mechanism (how the system does it)

In principle, each signal class may possess its own effective regulatory configuration. However, the present paper does not propose a full per-channel parameterization of SRF constructs. Such a model would produce a high-dimensional tensor (8 channels × 5 parameters × 2 states) that exceeds current theoretical justification. Instead, the present framework proposes only that effective regulatory parameters vary across signal classes — without specifying the full parameter set for each class. The degree and nature of this variation is an empirical question for future investigation.

6.3 What This Adds

SRF alone predicts that a system with low IRC and low SF will show compressed processing. Signal-Class Architecture adds: compressed processing of *what*? The answer is not "everything" — it is specific signal classes determined by the system's structural organization.

6.4 Scope Constraint

The compression profile describes processing topology — where representational resolution is high and where it is compressed. It does not claim to constitute the entirety of personality architecture.

Personality organization within the broader SRF framework involves multiple layers:

Layer	What it describes	Paper
Regulatory mechanism	How regulation operates (IRC, SF, ERD, REL, EV)	Paper 1
Processing topology	Where resolution varies across signal classes	Paper 4 (this paper)
Vulnerability topology	What becomes threatening and why	Paper 5

Layer	What it describes	Paper
Change dynamics	Why architecture resists modification	Paper 2
Stake depth	How deep the regulatory target goes	Paper 6

The compression profile is one layer — the observable processing surface. It does not replace self-architecture (which explains *why* specific signals compress), regulatory mechanism (which explains *how* compression is maintained), or change dynamics (which explains *why* compression persists). Personality is the full stack, not any single layer.

This constraint is important because it prevents the compression profile from becoming an overextended explanatory construct. It describes pattern, not cause. It maps topology, not motivation.

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7. Illustrative Application

To demonstrate explanatory utility, a brief illustrative application to narcissistic organization (described more fully in Halmetoja, 2026c) is offered.

A narcissistic regulatory organization may exhibit a characteristic compression profile in which status and control channels operate at high resolution while shame, dependency, rejection, and separateness channels are severely compressed. This single structural description explains several clinically recognized features without requiring global dysfunction: competence coexisting with relational primitiveness, empathy collapse under self-threat (potentially through reduced processing bandwidth when shame or rejection co-activates), categorical responses to rejection, and preserved strategic functioning.

This application is illustrative. Not all narcissistic presentations share identical compression profiles. The framework proposes family resemblances in compression patterns rather than categorical uniformity.

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8. Relationship to Existing Theory

The present framework must be explicitly distinguished from related but non-identical theoretical traditions.

8.1 Trait Psychology

Trait models (Big Five, HEXACO) describe broad dispositional tendencies assumed to operate relatively consistently across contexts. The present framework proposes that regulatory architecture is signal-class-specific — the same individual may show different effective trait-like properties depending on which signal class is activated. This is not inconsistency but structure.

8.2 Mentalization

Mentalization (Fonagy et al., 2002) describes the capacity to understand behavior in terms of mental states. The present framework proposes that mentalization capacity may itself be signal-class-dependent — a person may mentalize effectively regarding others' professional motivations while failing to mentalize regarding attachment-relevant intentions. This is consistent with Fonagy's observation that mentalization degrades under attachment activation, but extends it to a broader signal-class architecture.

8.3 Emotion Regulation

Emotion regulation research (Gross, 2015) describes processes by which individuals manage emotional experience. The present framework does not describe management processes but resolution architecture — the structural capacity to differentiate signals before regulation strategies are applied. Compression may constrain or shape subsequent regulation strategy selection.

8.4 Attachment Theory

Attachment theory (Bowlby, 1969; Mikulincer & Shaver, 2007) explains relational threat dynamics and their developmental origins. The present framework generalizes beyond attachment to a broader signal-class architecture. Attachment is one signal class among several — important but not exhaustive of the regulatory topology.

8.5 Rejection Sensitivity

Rejection sensitivity (Downey & Feldman, 1996) describes heightened expectation and reactivity to rejection cues. The present framework reinterprets this as low representational resolution in the rejection signal class — ambiguous social signals are compressed into categorical rejection interpretations because the system cannot sustain differentiated processing of rejection-relevant information.

8.6 Schema Modes

Schema therapy (Young et al., 2003) proposes that individuals shift between distinct schema modes under activation. The present framework shares the observation of state-dependent reorganization but differs structurally: schema modes describe holistic experiential states, while compression profiles describe signal-class-specific resolution variation within a single state. A person need not shift into a different "mode" to show differential resolution — they may simultaneously process status at high resolution and shame at low resolution within the same experiential moment.

8.7 Context-Dependent Mentalization Collapse

Luyten and Fonagy (2015) describe mentalization as context-sensitive, degrading under attachment activation. The present framework is consistent with this observation but proposes a more general architecture: degradation is not limited to mentalization, not limited to attachment contexts, and not limited to a single dimension. It is a multi-channel phenomenon affecting representational resolution across distinct signal classes, each with its own degradation profile.

8.8 Distinguishing the Present Framework

The critical distinction from all of the above is architectural specificity. Existing frameworks typically describe either: - Global capacity that degrades under stress (mentalization, distress tolerance) - Domain-specific sensitivity that heightens reactivity (rejection sensitivity, attachment anxiety) - State shifts between holistic modes (schema modes) - Cognitive biases that distort processing (motivated reasoning, cognitive dissonance) - Appraisal processes that evaluate threat (appraisal theory) - Granularity of emotional experience (emotional granularity / differentiation) - Prediction-driven perception (predictive processing)

The present framework proposes something structurally different from each:

Existing approach	What it explains	How SRF differs
Motivated reasoning	Why people resist disconfirming evidence	SRF specifies *which* evidence resists — determined by signal-class architecture, not general motivation
Appraisal theory	How events are evaluated as threatening	SRF proposes that appraisal resolution itself varies structurally across signal classes — not just the outcome but the representational depth of the appraisal
Emotional granularity	Individual differences in emotion differentiation	SRF proposes that granularity is signal-class-specific within individuals — not a global trait but a structured topology
Predictive processing	How priors shape perception	SRF specifies *which* priors become rigid and why — linking prediction rigidity to self-architecture vulnerability rather than general Bayesian updating

The unit of analysis is not the global capacity, not the single sensitive domain, not the holistic state, not the cognitive bias, and not the general prediction system — it is the multi-channel compression profile as a structural feature of personality organization.

8.9 Summary of Novelty Claim

To state the novelty claim as directly as possible: existing frameworks explain *that* processing varies (emotional granularity), *that* it degrades under threat (mentalization), *that* people resist disconfirming evidence (motivated reasoning), or *that* priors shape perception (predictive processing). The present framework proposes *where* processing varies within a single individual, *which specific signals* degrade and which are preserved, and *why* the pattern takes the specific form it does — linking observable compression topology to underlying self-architecture vulnerability. No existing framework provides this three-layer integration (mechanism + topology + vulnerability) within a single structural model.

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9. Testable Predictions

9.1 Within-Subject Resolution Variation

The same individual should show measurably different representational resolution across signal classes. This can be tested through within-subject designs comparing processing sophistication (ambiguity tolerance, interpretive complexity, affective differentiation) across signal-class-specific activation conditions.

9.2 Stress-Selective Degradation

Stress should selectively degrade resolution in vulnerable channels while relatively preserving resolution in non-vulnerable channels. This predicts an interaction between stress and signal class — not a main effect of stress alone.

9.3 Characteristic Clinical Signatures

Recognizable clinical presentations (narcissistic, avoidant, anxious, borderline) should show characteristic compression profile signatures — not identical profiles within each category, but family resemblances in which signal classes are most compressed.

9.4 Channel-Specific Empathy Failure

Empathy failure may emerge from channel-specific bandwidth collapse rather than global empathic incapacity. A person may demonstrate preserved empathy for others' professional frustrations while showing empathy failure for others' attachment needs — because the attachment signal class is compressed in their own system, reducing available resolution for processing attachment-relevant information from others.

9.5 Independence from Cognitive Ability

Compression profiles should be independent of general cognitive ability. Intelligence should not predict which signal classes are compressed or the degree of compression under activation.

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10. Toward Operationalization

10.1 Signal-Class-Specific Resolution Measurement

Operationalizing representational resolution requires signal-class-specific paradigms:

- **Ambiguity tolerance tasks**: presenting ambiguous scenarios within specific signal classes and measuring interpretive complexity of responses
- **Forced differentiation tasks**: requiring participants to generate multiple interpretations of signal-class-specific stimuli
- **Narrative complexity coding**: analyzing spontaneous narratives for differentiation level within specific signal domains
- **Ecological momentary assessment**: sampling real-time processing sophistication across naturally occurring signal-class activations

10.2 Compression Profile Assessment

A compression profile assessment would require:

1. Baseline resolution measurement across all canonical signal classes
2. Stress-condition resolution measurement (signal-class-specific activation)
3. Computation of within-subject resolution variation and degradation patterns

10.3 Operational Caveat

No validated compression profile instrument currently exists. The measurement proposals above are conceptual directions, not established paradigms. The gap between the framework's theoretical specificity and its current measurement status is substantial.

Specifically: - No signal-class-specific resolution measure has been developed or validated - No within-subject compression profile assessment protocol exists - The distinction between compression and other forms of processing degradation (cognitive load, fatigue, alexithymia) has not been empirically established - Whether compression profiles show the temporal stability required for personality-level description remains unknown

The present paper's scientific value therefore lies in generating testable structural predictions — not in claiming current assessability. The framework is empirically approachable in principle but not yet empirically realized in practice. Instrument development represents the highest practical priority for the broader research program.

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11. Limitations

Several limitations should be explicitly acknowledged.

1. **Conceptual framework only.** No empirical validation is presented.
2. **Provisional signal taxonomy.** The eight proposed signal classes are derived from existing literatures but have not been empirically validated as a coherent taxonomy. Factor-analytic investigation may reveal different or additional classes.
3. **Construct overlap.** Representational resolution overlaps conceptually with mentalization, cognitive complexity, ambiguity tolerance, and emotional differentiation. The framework must demonstrate incremental explanatory value beyond these existing constructs.
4. **Metaphorical resolution language.** "Resolution" and "compression" are structural analogies, not literal descriptions of neural computation. This should not be mistaken for a computational model.
5. **No validated instrument.** Compression profile assessment does not yet exist as a psychometric tool.
6. **Causal mechanisms inherited from SRF.** The present framework explains where compression occurs but relies on SRF for mechanistic explanation of why. If SRF constructs prove empirically problematic, the present framework's explanatory power is correspondingly reduced.
7. **Risk of proliferating constructs.** Eight signal classes, each with baseline and stress resolution levels, produces a high-dimensional description. Whether this complexity is justified by explanatory gain or represents unnecessary proliferation remains an empirical question.

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12. Summary

The present paper proposes that psychological regulation is signal-class-dependent rather than globally scalar. Different psychologically meaningful signal classes — attachment, dependency, rejection, separateness, shame, status, control, intent attribution — may sustain different effective representational resolution levels, producing characteristic compression profiles that explain paradoxical functioning without requiring global dysfunction.

The framework's contribution is not the trivial observation that people process things differently. It is the proposal of a formal structural model of differential representational resolution across canonical psychologically meaningful signal classes, dynamically modulated by activation state and linked to regulatory architecture.

If Paper 1 (SRF) was the engine paper — explaining how regulation works — the present paper is the topology paper — explaining where reality compresses.

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